

Thomas J. Lane

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GitHub: github.com/tjlane

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Group Site: pbio.cfel.de

Personal Site: tjlane.github.io

Education

Stanford University, Stanford CA. PhD Chemistry, April 2015.

Advisor: Vijay Pande. Thesis: "Modern theory of protein folding."

Pomona College, Claremont CA. BA Chemistry, May 2010.

Academic Positions

Group Leader, Deutsches Elektronen-Synchrotron "DESY"

Hamburg, DE. Aug/2020-present

Leader: Photobiology Group. Center for Free-Electron Laser Science (CFEL).

Co-Founder, Reciprocal Space Station "RSS"

NYC, USA. Jan/2026-present

Academic consortium for open-source structural biology software.

Staff Scientist, SLAC Nat'l Accelerator Lab

Menlo Park, CA. Jan/2020-July/2020

LCLS & Machine Learning group. Permanent academic position.

Associate Staff Scientist, SLAC Nat'l Accelerator Lab

Menlo Park, CA. June/2015-Jan/2020

Linac Coherent Light Source (LCLS). Tenure-track.

Visiting Positions

Visiting Associate Research Scientist, Columbia University

New York, NY. Oct/2025-present

Sabbatical in the laboratory of Mohammed AlQuraishi.

Visiting Scholar, Stockholm University

Stockholm, SE. Mar/2019-July/2019

Visiting scientist in the laboratory of Anders Nilsson.

Industry Experience

Principal Scientist, CHARM Therapeutics

Cambridge, UK. Nov/2022-May/2024

Technical strategy & ML/experiment intersection.

Scholarship

Publications: 75 of which PI role in project: 12

of which *Science* or *Nature*: 6 of which *arXiv* preprints: 3

Total Citations: 10,500+ H-index: 40

Book Chapters: 1 Scholarly Grants & Awards: 13

→ <https://scholar.google.com/citations?user=fWw8K2gAAAAJ&hl=en>

Publications

Articles: Corresponding Author Following PI Role

12. Creon, A.; Scheer, T. E. S.; Reinke, P.; Rahmani, A.; Günther, S.; Niebling, S.; Schamoni-Kast, K.; Uetrecht, C.; Meents, A.; Chapman, H. N.; Sprenger, J.; **Lane, T. J.** (2026). Statistical crystallography reveals an allosteric network in SARS-CoV-2 M^{PRO}. *Commun Biol* 9, 602.
11. Fadini, A.; Apostolopoulou, V.; **Lane, T. J.**; van Thor, J. J. (2025). Denoising and iterative phase recovery reveal low-occupancy populations in protein crystals. *Commun Biol* 8, 1649.
10. Stransky, M.; **Lane, T. J.**; Gorel, A.; Boutet, S.; Schlichting, I.; Mancuso, A. P.; Jurek, Z.; Ziaja, B. (2024). Ionization by XFEL radiation produces distinct structure in liquid water. *Commun Phys* 7, 281.
9. Reinke, P. Y. A.; Schubert, R.; et al.; **Lane, T. J.** (2024). SARS-CoV-2 M^{PRO} responds to oxidation by forming disulfide and NOS/SONOS bonds. *Nat Commun* 15, 3827.
8. Christou, N-E; Apostolopoulou, V; Melo, D. V. M.; et al.; **Lane, T. J.** (2023). Time-resolved crystallography captures light-driven DNA repair. *Science* 382, 1015-1020.
7. **Lane, T. J.** (2023). Protein structure prediction has reached the single-structure frontier. *Nat Methods* 20, 170-173.
6. **Lane, T. J.**; Ratner, D. (2020). What are the advantages of ghost imaging? Multiplexing for x-ray and electron imaging. *Opt Express* 28, 5898-5918.
5. Barmherzig, D.; Sun, J.; Li, P.-N.; **Lane, T. J.** (2018). On Block-Reference Coherent Diffraction Imaging. *Imaging and Applied Optics* 10.1364/COSI.2018.CTH1B.1.
4. **Lane, T. J.** (2018). Biomolecular Imaging with Megavolt Electrons: Opportunities and Challenges for an Ultrafast Electron Microscope at SLAC. *SLAC SciDocs* SLAC-R-1093.
3. Peck, A.; Poitevin, F.; **Lane, T. J.** (2018). Intermolecular correlations are necessary to explain diffuse scattering from protein crystals. *IUCrJ* 5, 211-222.
2. Shi, J.; Nobrega, R. P.; Schwantes, C.; Kathuria, S. V.; Bilsel, O.; Matthews, C. R.; **Lane, T. J.**; Pande, V. S. (2017). Atomistic structural ensemble refinement reveals non-native structure stabilizes a sub-millisecond folding intermediate of CheY. *Sci Rep* 7, 44116.
1. Shi, L.; Wetzstein, G.; **Lane, T. J.** (2016). A Flexible Phase Retrieval Framework for Flux-limited Coherent X-Ray Imaging. *arXiv:1606.01195*.

Articles: First Author

9. **Lane, T. J.**; Schwantes, C. R.; Beauchamp, K. A.; Pande, V. S. (2014). Efficient inference of protein structural ensembles. *arXiv:1408.0255*.
8. **Lane, T. J.**; Schwantes, C. R.; Beauchamp, K. A.; Pande, V. S. (2013). Probing the Origins of Two-State Folding. *J Chem Phys* 139, 145104.
7. **Lane, T. J.**; Pande, V. S. (2013). Inferring the Rate-Length Law of Protein Folding. *PLoS ONE* 8, e78606.
6. **Lane, T. J.**; Shukla, D.; Beauchamp, K. A.; Pande, V. S. (2013). To milliseconds and beyond: challenges in the simulation of protein folding. *Curr Opin Struct Biol* 23, 58-65.

5. **Lane, T. J.**; Pande, V. S. (2012). Eigenvalues of the Homogeneous Finite Linear One Step Master Equation: Applications to Downhill Folding. *J Chem Phys* 137, 215106.
4. **Lane, T. J.**; Pande, V. S. (2012). A Simple Model Predicts Experimental Folding Rates and a Hub-Like Topology. *J Phys Chem B* 116, 6764-6774.
3. **Lane, T. J.**; Bowman, G. R.; Beauchamp, K. A.; Voelz, V. A.; Pande, V. S. (2011). Markov State Model Reveals Folding and Functional Dynamics in Ultra-Long Trajectories. *J Am Chem Soc* 133, 18413-18419.
2. **Lane, T. J. et al.** (2008). Dual-Beam Polarization Interferometry Resolves Mechanistic Aspects of Polyelectrolyte Adsorption. *Langmuir* 24, 10633-10636.
1. **Lane, T. J. et al.** (2008). Ultrasonic Rheology of the Aqueous Phase of SDS Micellar Systems. *Anal Chem* 80, 7840-7845.

Articles: Co-authored

54. Schori, A. *et al.* (2025). Real-space observation of the dissociation of a transition metal complex and its concurrent energy redistribution. *Nat Commun* 16, 4767.
53. Ma, L. *et al.* (2025). Revealing the reaction path of UVC bond rupture in cyclic disulfides with ultrafast x-ray scattering. *Sci Adv* 11, eadp9175.
52. Fletcher, L. *et al.* (2024). X-ray diffraction of metastable structures from supercooled liquid hydrogen. *Sci Rep* 14, 17283.
51. Peck, A.; **Lane, T. J.**; Poitevin, F. (2023). Modeling diffuse scattering with simple, physically interpretable models. *Methods Enzymol* 688, 169-194.
50. Aithani, L. *et al.* (2023). Advancing structural biology through breakthroughs in AI. *Curr Opin Struct Biol* 80, 102601.
49. Amann-Winkel, K. *et al.* (2023). Liquid-liquid phase separation in supercooled water from ultrafast heating of low-density amorphous ice. *Nat Commun* 14, 442.
48. Yang, C. *et al.* (2023). Melting domain size and recrystallization dynamics of ice revealed by time-resolved x-ray scattering. *Nat Commun* 14, 3313.
47. Al-Haddad, A., *et al.* (2022). Observation of site-selective chemical bond changes via ultrafast chemical shifts. *Nat Commun* 13, 7170.
46. Chitturi, S. R. *et al.* (2022). A machine learning photon detection algorithm for coherent x-ray ultrafast fluctuation analysis. *Structural Dynamics* 9, 054302.
45. Hammarström, B.; **Lane, T. J.**; Batili, H.; Sierra, R.; Wiklund, M.; Sellberg, J. A. (2022). Acoustic Focusing of Protein Crystals for In-Line Monitoring and Up-Concentration during Serial Crystallography. *Anal Chem* 94, 12645-12656.
44. Adam, V. *et al.* (2022). Rational Control of Off-State Heterogeneity in a Photoswitchable Fluorescent Protein Provides Switching Contrast Enhancement. *ChemPhysChem*, e202200192.
43. Srinivasan, V. *et al.* (2022). Antiviral activity of natural phenolic compounds in complex at an allosteric site of SARS-CoV-2 papain-like protease. *Commun Biol* 5, 805.

42. Ladd-Parada, M. *et al.* (2022). Following the Crystallization of Amorphous Ice after Ultrafast Laser Heating. *J Phys Chem B* 126, 2299.
41. Esmaeildoost, N. *et al.* (2021). Anomalous temperature dependence of the experimental x-ray structure factor of supercooled water. *J Chem Phys* 155, 214501.
40. Sun, Y. *et al.* (2021). Nonuniform Flow Dynamics Probed by Nanosecond X-Ray Speckle Visibility Spectroscopy. *Phys Rev Lett* 127, 058001.
39. Sorigue, D. *et al.* (2021). Mechanism and dynamics of fatty acid photodecarboxylase. *Science* 372, eabd5687.
38. Günther, S.; Reinke, P. Y. A.; *et al.* (2021). X-ray screening identifies active site and allosteric inhibitors of SARS-CoV-2 main protease. *Science* 372, 642-646.
37. Yun, J.-H. *et al.* (2021). Early-stage dynamics of chloride ion-pumping rhodopsin revealed by a femtosecond X-ray laser. *PNAS* 118, e2020486118.
36. Pathak, H. *et al.* (2021). Enhancement and maximum in the isobaric specific-heat capacity measurements of deeply supercooled water using ultrafast calorimetry. *PNAS* 118, e2018379118.
35. Gruenbein, M. L. *et al.* (2021). Observation of shock-induced protein crystal damage during megahertz serial femtosecond crystallography. *Phys. Rev. Research* 3, 013046.
34. Coakley J. *et al.* (2020). Femtosecond quantification of void evolution during rapid material failure. *Sci Advances* EABB4434.
33. Kim, K.-H.; Amann-Winkel, K. *et al.* (2020). Experimental observation of the liquid-liquid transition in bulk supercooled water under pressure. *Science* 370, 978-982.
32. Kabra, K.; Li, S.; Cropp, F.; **Lane, T. J.**; Musumeci, P.; Ratner, D. (2020). Mapping photocathode quantum efficiency with ghost imaging. *Phys Rev Accel Beams* 23, 022803.
31. Driver, T. *et al.* (2020). Attosecond Transient Absorption Spooktroscopy: a ghost imaging approach to ultrafast absorption spectroscopy. *Phys Chem Chem Phys* 22, 2704-2712.
30. Stankus, B. *et al.* (2019). Ultrafast X-ray scattering reveals vibrational coherence following Rydberg excitation. *Nat Chem* 11, 716-721.
29. Lieske, J. *et al.* (2019). On-chip crystallization for serial crystallography experiments and on-chip ligand-binding studies. *IUCrJ* 6, 714-728.
28. Barmherzig, D.; Sun, J.; Li, P.-N.; **Lane, T. J.**; Candes, E. J. (2019). Holographic Phase Retrieval and Reference Design. *Inverse Problems* 35, 094001.
27. Ratner, D.; Cryan, J. P.; **Lane, T. J.**; Li, S.; Stupakov, G. (2019). Pump-Probe Ghost Imaging with SASE FELs. *Phys Rev X* 9, 011045.
26. Ruddock, J. *et al.* (2019). Simplicity beneath Complexity: Counting Molecular Electrons Reveals Transients and Kinetics of Photodissociation Reactions. *Angew Chem* 58, 6371-6375.
25. Sierra, R. G.; Batyuk, A.; Sun, Z.; Aquila, A.; Hunter, M. S.; **Lane, T. J.**; Liang, M. *et al.* (2019). The Macromolecular Femtosecond Crystallography Instrument at the Linac Coherent Light Source. *J Synch Rad* 26, 346-357.
24. Barmherzig, D.; Sun, J.; Candes, E. J.; **Lane, T. J.**; Li, P.-N. (2019). Dual-Reference Design for Holographic Coherent Diffraction Imaging. *arXiv:1902.02492*.

23. Yong, H. *et al.* (2018). Determining Orientations of Optical Transition Dipole Moments Using Ultrafast X-ray Scattering. *J Phys Chem Lett* 9, 6556-6562.
22. Li, S.; Cropp, F.; Kabra, K.; **Lane, T. J.**; Wetzstein, G.; Musumeci, P.; Ratner, D. (2018). Electron ghost imaging. *Phys Rev Lett* 121, 114801.
21. Nogly, P. *et al.* (2018). Retinal isomerization in bacteriorhodopsin captured by a femtosecond x-ray laser. *Science* 361, eaat0094.
20. Perakis, F.; Camisasca, G.; **Lane, T. J.**; *et al.* (2018). Coherent x-rays reveal the influence of cage effects on ultrafast water dynamics. *Nat Commun* 9, 1917.
19. Coquelle, N. *et al.* (2018). Chromophore twisting in the excited state of a photoswitchable fluorescent protein captured by time-resolved serial femtosecond crystallography. *Nat Chem* 10, 31-37.
18. Shen, X.; Li, R. K.; Lundström, U.; **Lane, T. J.**; Reid, A. H.; Weathersby, S. P.; Wang, X. J. (2017). Femtosecond mega-electron-volt electron microdiffraction. *Ultramicroscopy* 184 pt. A, 172-176.
17. Herrmann, J. *et al.* (2017). Environmental Calcium Controls Alternate Physical States of the Caulobacter Surface Layer. *Biophys J* 112, 1841-1851.
16. Thayer, J. *et al.* (2017). Data systems for the Linac coherent light source. *Adv Struct Chem Imaging* 3, 3.
15. Stankus, B.; Budarz, J. M.; Kirrander, A.; Rogers, D.; Robinson, J.; **Lane, T. J.**; *et al.* (2016). Femtosecond photodissociation dynamics of 1,4-diiodobenzene by gas-phase X-ray scattering and photoelectron spectroscopy. *Faraday Discuss* 194, 525-536.
14. Young, I. D. *et al.* (2016). Structure of photosystem II and substrate binding at room temperature. *Nature* 540, 453-457.
13. Mendez, D. *et al.* (2016). Angular correlations of photons from solution diffraction at a free-electron laser encode molecular structure. *IUCrJ* 3, 420-429.
12. Hunter, M. S. *et al.* (2016). Selenium single-wavelength anomalous diffraction *de novo* phasing using an X-ray Free Electron Laser. *Nat Commun* 7, 13388.
11. Stan, C. A.; Milathianaki, D.; Laksmono, H.; Sierra, R. G.; McQueen, T. A.; Messerschmidt, M.; Williams, G. J.; Koglin, J. E.; **Lane, T. J.** *et al.* (2016). Liquid explosions induced by X-ray laser pulses. *Nat Phys* 12, 966-971.
10. Damiani, D.; Dubrovin, M.; Gaponenko, I.; Kroeger, W.; **Lane, T. J.**; Mitra, A.; O'Grady, C.P.; Salnikov, A.; Sanchez-Gonzalez, A.; Schneider, D.; Yoon, C. H. (2016). Linac Coherent Light Source data analysis using psana. *J Appl Cryst* 49, 672-679.
9. Mariani, V.; Morgan, A.; Yoon, C. H.; **Lane, T. J.**; White, T. A.; O'Grady, C.; *et al.* (2016). OnDA: online data analysis and feedback for serial X-ray imaging. *J Appl Cryst* 49, 1073-1080.
8. Minitti, M. P.; Budarz, J. M.; Kirrander, A.; Robinson, J. S.; Ratner, D.; **Lane, T. J.**; Zhu, D.; Glowacki, J.M.; Kozina, M.; Lemke, H. T.; Sikorski, M.; Feng, Y.; Nelson S.; Saita, K.; Stankus B.; Northey T.; Hastings, J. B.; Weber, P. M. (2015). Imaging Molecular Motion: Femtosecond X-Ray Scattering of an Electrocyclic Chemical Reaction. *Phys Rev Lett* 114, 255501.
7. McGibbon, R. T.; Beauchamp, K. A.; Schwantes, C. R.; Wang, L.-P.; Hernandez, C. X.; Harrigan, M. P.; **Lane, T. J.**; Swails, J. M.; Pande, V. S. (2015). MDTraj: a modern, open library for the analysis of molecular dynamics trajectories. *Biophys J* 109, 1528-1532.

6. Mendez, D.; **Lane, T. J.**; *et al.* (2014). Observation of correlated X-ray scattering at atomic resolution. *Phil Trans R Soc B* 369, 20130315.
5. Eastman, P.; Friedrichs, M. S.; Chodera, J. C.; Radmer, R. J.; Bruns, C. M.; Ku, J. P.; Beauchamp, K. A.; **Lane, T. J.**; Wang, L.-P.; Shukla, D.; Tye, T.; Houston, M.; Stich, T.; Klein, C.; Shirts, M. R.; Pande, V. S. (2013). OpenMM 4: A Reusable, Extensible, Hardware Independent Library for High Performance Molecular Simulation. *J Chem Theory Comput*, 9, 461-469.
4. Beauchamp, K. A.; Bowman, G. R.; **Lane, T. J.**; Maibaum, L.; Haque, I. S.; Pande, V. S. (2011). MSMBuild2: Modeling Conformational Dynamics on the Picosecond to Millisecond Scale. *J Chem Theory Comput* 7, 3412-3419.
3. Frau, A. F.; **Lane, T. J.**; Schlather, A. E.; Park, J. Y.; Advincula, R. C. (2011). Modulating Electrochemical Activity in Polyaniline/Titanium Oxide Hybrid Nanostructured Ultrathin Films. *Ind Eng Chem Res* 50, 5532-5542.
2. Zwang, T. J.; Fletcher, W. R.; **Lane, T. J.**; Johal, M. S. (2010). Quantification of the Layer of Hydration of a Supported Lipid Bilayer. *Langmuir* 26, 4598-4601.
1. Kett, P. J.; Casford, M. T.; Yang, A. Y.; **Lane, T. J.**; Johal, M. S.; Davies, P. B. (2009). Structural Changes in a Polyelectrolyte Multilayer Assembly Investigated by Reflection Absorption Infrared Spectroscopy and Sum Frequency Generation Spectroscopy. *J Phys Chem B* 113, 1559-1568.

Book Chapters

Ultrasonic Rheology of Mixed-Phase Systems: Using a QCM as an Effective Rheological Device: Theory and Applications for Combined Surface and Bulk Rheology. **Thomas J. Lane**. *Amphiphiles: Molecular Assembly and Applications*. Ed. Ramanathan Nagarajan. ACS Symposium Series 1070, Chapter 10, pp 145-174 (2011).

Service

Co-founder, The Reciprocal Space Station (<https://rs-station.org>).

Co-editor, Journal of Applied Crystallography (IUCr).

Notable Grants, Awards, Honors

Norddeutscher Wissenschaftspreis (North German Science Prize) “Magnetsinn von Zugvögeln” collaboration (U. Hamburg, U. Oldenburg, DESY, EuXFEL, 2026).

Royal Society of Chemistry Horizon Award: Ultrafast X-ray Scattering (2021).

Faculty, International Max Planck Research School for Ultrafast Imaging & Structural Dynamics (2021).

Member, DFG Excellence Cluster on Advanced Imaging of Matter “AIM” (2020).

Helmholtz Young Investigator Group Leader Award, 1.8M EUR (2020-2026).

DOE SciDAC: Designing Photocatalysts Through Scalable Quantum Mechanics and Dynamics. Principal author with T. Martinez, \$4M. Managed 4-PI photocatalyst design team (2017-2020).

NSF Graduate Research Fellowship (2010-2013).

John Stauffer Academic Award (2010).
The Frank Parkhurst Brackett, Jr. and David Wark Brackett Prize (2010).
Wig Grant for Curriculum Development (2010).
Undergraduate Award in Analytical Chemistry (2009).
Barry M. Goldwater Scholarship (2009).
Beckman Scholarship (2008).

Invited Talks

Fighting quantum yield: two challenging SFX targets. Science@FELs, Paul Scherrer Institute, Villigen (2026).
Life beyond the single structure frontier. Uppsala University, Uppsala (2026).
Life at the atomic scale - filming photoactive proteins. Carl von Ossietzky University, Oldenburg (2025).
Time-resolved crystallography captures light-driven DNA repair. Photon Science Seminar, SLAC (2025).
Time-resolved crystallography captures light-driven DNA repair. EBSA, Rome (2025).
Time-resolved crystallography captures light-driven DNA repair. EMBO Workshop: Time-resolved spectroscopy meets time-resolved crystallography, Pécs (2025).
Time-resolved crystallography captures light-driven DNA repair. U. Düsseldorf Chemistry Colloquium, Düsseldorf (2025).
Time-resolved crystallography captures light-driven DNA repair. Ultrafast Dynamic Imaging of Matter Conference, Hamburg (2024).
Time-resolved crystallography captures light-driven DNA repair. UKXFEL Meeting. Royal Society, London (2024).
Time-resolved crystallography captures light-driven DNA repair. 7th Ringberg Workshop on Structural Biology. Tegernsee (2024).
Statistical crystallography reveals correlated motions of SARS-CoV-2 M^{Pro}. CFEL retreat, Timmendorf (2022).
Can we witness proteins traversing conical intersections via FEL-based crystallography? European Crystallographic Meeting, Versailles (2022).
Statistical crystallography reveals correlated motions of SARS-CoV-2 M^{Pro}. Pacificchem (online) (2021).
Statistical crystallography reveals correlated motions of SARS-CoV-2 M^{Pro}. Helmholtz: Matter, Materials, Life (MML) Workshop (online) (2021).
Revealing allosteric pathways via large-scale crystallographic studies. BioXFEL Meeting (online) (2021).
The quest for correlated motion in proteins: progress through diffuse scatter and multi-crystal analysis. CUI Seminar, Hamburg (2020).
What Can We Learn from Diffuse Scatter? Applied Physics Seminar, KTH Stockholm (2019).
Putting the Femtosecond in SFX: Ultrafast Crystallography. Pittsburgh Diffraction Meeting (2018).

What Can We Learn from Diffuse Scatter? 5th Ringberg Conference on XFELs in Biology (2018).

Phase Retrieval with Prior Information. LCLS/SSRL Users' Meeting, Menlo Park, CA (2016).

Protein Evolution and Design: Session Chair. Protein Folding GRC, Galveston, TX (2014).

Boltzmann Takes On Biology: The Next Frontier in Structure. Chemistry Colloquium, Pomona College, CA (2013).

Transition State Ensembles: The View from Simulation and Experiment. Protein Folding Consortium, Stony Brook, NY (2012).

To the Exa-Scale and Beyond: Statistically Motivated Linear Scaling for Protein Dynamics. Titan Summit, Oak Ridge, TN (2011).

Markov State Model Reveals Folding and Functional Dynamics in Ultra-Long Trajectories. Biophysics Symposium, Stanford, CA (2010).

Ultrasonic Rheology of Mixed-Phase Systems. Amgen, Thousand Oaks, CA (2010).

Ultrasonic Rheology of Surfactant Systems. Biolin Scientific Surface Science Symposium, Pomona College, CA (2010).

Ultrasonic Rheology of SDS Microemulsions. Stanford QCM-D Users' Meeting. *Best Presentation Award* (2007).